



ARTIFICIAL INTELLIGENCE



Fig. 4: RT-RK Alpha board for autonomous vehicle solution
(Credit: www.rt-rk.com)

virtual neural network of self-driving vehicles that learn as they go.

Infotainment systems. In-vehicle infotainment provides a combination of information and entertainment in the form of audio and video content, games, social networking, etc. Infotainment is a part of the technological interface of any modern car. AI improves infotainment systems in a better way. Information related to driver assistance, navigation, access to phone calls, messaging and the like is delivered seamlessly to the driver through infotainment systems.

Features like eye tracking, speech recognition, gesture recognition, monitoring driving and a database of natural languages are included in AI systems. Moreover, with AI's connection to Big Data, infotainment systems can be used to suggest products and services to drivers.

Israeli automotive computer vision startup eyeSight has developed a program that recognises simple hand gestures to control the infotainment system without taking its eyes off the road.

AI driver assistance

AI can do more than autonomous driving, that is for sure! It keeps an eye on the human driver and ensures safe driving. AI software detects whether or not the driver is in the vehicle. AI monitors various sensors installed in the vehicle. A connected vehicle can check engine lights, oil lights and low-battery indicators, and detect problems before these affect vehicle operation. By monitoring thousands of sensors, AI can spot minute changes and component failures.

AI can identify dangerous situations and alert the driver, or take

emergency control of the vehicle, to avoid an accident. Emergency braking, cross-traffic detection, blind spot monitoring and driver-assist steering can help avoid accidents, and save lives in the process.

eyeSight uses AI and deep learning to offer a plethora of in-car automotive solutions. Using advanced time-of-flight

cameras and IR sensors, eyeSight's AI software can detect many important key areas like facial recognition, eye tracking, upper-body detection and so on.

Facial recognition. These algorithms detect which driver is operating the vehicle. A husband, wife or a young adult can have his or her own preferences. The system can automatically adjust the seat, mirrors and temperature to suit the individual.

Eye tracking. On observing eye gaze, eye openness and head position, eyeSight can detect distracted driving and alert the driver to keep the eyes on the road. Drowsiness is detected by eye openness and head position, allowing the system to wake the driver up, if necessary.

Contextual control. The system tailors the content of the heads-up display (HUD) according to where the driver's eyes are focused.

Upper-body detection. It detects the driver's posture. Air bags can be deployed based on posture at the time of crash, reducing injuries.

AI automotive insurance

The insurance industry supports the use of AI in cars to help make risk assessments in real time. AI speeds up the process of filing claims when accidents do occur. Ant Financial has released its AI-powered video app in the Chinese auto market. The app enables drivers to perform their own auto damage assessment for their respective insurance companies.

On-screen instructions guide users to record video of their vehicle damage for insurance claim. AI quickly informs the drivers how to have their vehicle repaired, and what will be covered by insurance.

AI in automobile manufacturing

AI is changing what vehicles can do and how these are built. Assembly-line robots are nothing new. But now, AI-powered smart robots can work with their human counterparts. Collaborative robots use AI to sense what human workers are doing, and they adjust their motions to avoid injuring human co-workers.

Kia Motors has started developing wearable industrial robots for assembly lines. Wearable robots help protect workers' knees, back and neck, while giving them the mobility and strength to perform tenuous jobs.

Automated guided vehicles are being used to move materials around automotive plants without human intervention. AI enables these delivery vehicles to identify objects in their paths and to adjust their route, accordingly.

Painting and welding robots, when powered by AI, can do more than follow a pre-programmed routine. AI empowers them to identify defects or irregularities in materials and alert quality control personnel.

The way forward

AI has captured maximum attention in automotive applications, particularly in driverless cars. This technology has brought a new breed of car driving. Huge computational needs in driverless cars have taken AI to a completely different level. Advancements in AI have contributed to the growth of the automotive industry through the creation and evolution of such self-driving vehicles as cars, taxis, trucks, buses and bikes.

Once driverless technology is fully developed, traffic collisions and accidents caused by human error, aggressive driving and other inconveniences will reduce substantially. AI not only enables autonomous driving and ensures safety of human drivers, it also helps in the manufacturing of vehicles. AI is already making dramatic changes in the automobile industry, including the way we drive and the way we interact with connected devices. The world of automotive industry with AI is going towards a new transformation, in a big way. **EFY**